

Surface Form, Not Intent: What Jailbreak-Wrapped Benign Supervision Actually Teaches

Anonymous ACL submission

Abstract

Hard-example safety tuning can make a model robust by making it reluctant, and jailbreak-shaped benign counterexamples are used to repair the resulting overrefusal. It remains unclear whether such data teaches transferable *harmless intent*, jailbreak-specific adaptation, or merely sensitivity to added context. We introduce a same-core, same-target controlled training trial. For each harmless request, we construct a direct version (D), a jailbreak-wrapped version (W), and a tokenizer-length-matched neutral version (N'); the arms share target tokens, hard harmful examples, update positions, initialization, and update count, differing only in benign user context. Across a 24-seed primary Qwen2.5-72B recipe, an 8-seed alternative Qwen recipe, and a 24-seed Mistral-7B replication, W improves safe/harmful separation over D on *wrapped* test presentations but is nonpositive on the *identical direct cores*. With a qualified Llama-70B judge, the pooled direct effect is -0.322 ($p=.008$), the wrapped effect is $+0.435$ ($p=.014$), and their difference is $+0.758$ ($p=.0006$); a qualified Qwen-7B judge reproduces the sign pattern. Because judge agreement falls on wrapped prompts, we rescore the same 6,400 responses after removing the wrapper from the judge input. The wrapped W–D effect increases rather than disappears (Qwen paired change $+1.045$; Llama $+1.145$; both $p=.0001$), while an unchanged-XSTest placebo is near zero. Direct exposure of the judge to the wrapper therefore cannot explain the result. Two held-out wrapper families replicate the positive effect across nearly every primary-Qwen seed, although a frozen 2,400-item independent Llama audit is directionally concordant but underpowered. The evidence supports test-time surface-form dependence, not transferable intent or improved safety; human confirmation, independently powered cross-family corroboration, and a sharper direct anchor remain open.

1 Introduction

Safety-tuned language models are increasingly deployed as user-facing assistants, and a large body of alignment work tries to make them refuse harmful requests while remaining helpful (Bianchi et al., 2023; Wei et al., 2023; Jiang et al., 2024). A recurring tension is that the same interventions that suppress jailbreaks also make models refuse harmless prompts that merely *look* adversarial—exaggerated safety, or overrefusal (Röttger et al., 2023; Cui et al., 2024). A representative recent recipe, hard-example safety tuning (Gupta et al., 2026), mines the prompts the model most often fails, trains on the model’s own safe rollouts, and drives jailbreak attack success rates down. It also induces heavy overrefusal on jailbreak-shaped benign prompts, which it repairs by interleaving adversarially framed benign counterexamples. The stated hypothesis is that these counterexamples teach the model *harmless intent*.

This hypothesis is attractive but underdetermined. Adding jailbreak-wrapped benign examples can simultaneously change the benign source, exact core request, response target, order, and token length relative to ordinary benign data. Any of these factors could produce the observed reduction in overrefusal. More importantly, the intent hypothesis makes a strong, testable prediction: if the model learns that requests wearing jailbreak clothing can still be harmless, that abstraction should transfer to the identical requests presented without the wrapper. If instead the data teaches the model to comply when jailbreak surface cues are present, the benefit will be confined to wrapped presentations at test time. This would be a *surface-form dependence*, or shortcut (Geirhos et al., 2020; McCoy et al., 2019), rather than an intent abstraction. Recent evidence that refusal is governed by low-dimensional, surface-sensitive representations (Arditi et al., 2024; Addepalli et al., 2024;

Xue et al., 2026) makes this alternative concrete.

In this paper, we introduce a same-core, same-target controlled trial that isolates what jailbreak framing on otherwise identical benign supervision actually buys. For each harmless core, we construct a contrast-set triple (Gardner et al., 2020): a dewrapped core (D), the same core inside a real jailbreak wrapper (W), and a tokenizer-length-matched neutral frame (N'). All arms interleave identical hard harmful examples and share every audited response target, training-example position, initialization, and update count; they differ only in benign user context. We train these arms across a 24-seed primary Qwen2.5-3B recipe, an 8-seed alternative Qwen recipe, and a 24-seed Mistral-7B replication, then evaluate safe/harmful separation on wrapped and direct presentations with two independently qualified automatic judges.

We find a consistent dissociation. W improves separation over D on wrapped test presentations, but is nonpositive on the identical direct cores, and the wrapped-minus-direct moderation is positive under both judges. Benign-wrapper supervision therefore buys separation where a wrapper is present at test time, without evidence of transfer to the underlying direct request. Because the phenomenon under study is surface form, the central alternative explanation is that the judge itself reacts to wrapper text, a concern supported by documented LLM-as-a-judge vulnerabilities (Shi et al., 2024; Maloyan et al., 2025; Zheng et al., 2023). We address this with an evaluator-context ablation: re-scoring identical stored responses while hiding the wrapper from the judge increases rather than removes the wrapped effect. This rules out direct judge exposure to the wrapper as the sole cause, while showing that evaluator context still changes estimated magnitude.

Two wrapper families held out from training reproduce the positive wrapped effect over the full 24-seed primary-Qwen panel. A frozen independent Llama-70B audit agrees directionally but does not reach its preregistered per-family corroboration threshold. We consequently scope the result as multiple-automatic-judge-qualified and output-attributed, not human-corroborated or independently cross-family-locked. The W-N' contrast remains a judge-dependent exploratory signal, and no tested setting supports a safety benefit. We will release the triples, hashes, adapters, judge outputs, cross-family audit, and analysis code.

2 Related Work

Safety tuning and exaggerated safety. Adding safety data during instruction tuning improves refusal but can induce exaggerated safety, in which models reject harmless prompts that superficially resemble unsafe ones (Bianchi et al., 2023). XSTest (Röttger et al., 2023) and OR-Bench (Cui et al., 2024) operationalize this failure with safe/unsafe contrasts; SORRY-Bench (Xie et al., 2024) systematizes refusal measurement and evaluates scalable safety judges. Fine-tuning on even benign data can also erode prior alignment (Qi et al., 2023), which underscores why the source and framing of repair data must be controlled. The seed recipe (Gupta et al., 2026) removes jailbreaks through hard-example tuning, incurs overrefusal, and repairs it with adversarially framed benign counterexamples hypothesized to teach harmless intent. It does not hold benign source, core request, target, order, and length fixed between adversarial-benign and ordinary-benign conditions. We target this identification gap.

Surface form versus intent. Controlled minimal pairs distinguish genuine capabilities from surface heuristics in contrast sets (Gardner et al., 2020), syntactic diagnostics (McCoy et al., 2019), and shortcut-learning analyses (Geirhos et al., 2020). In safety, alignment can fail under mismatched presentations (Wei et al., 2023); semantically related natural prompts can reveal failures without being crafted attacks (Addepalli et al., 2024); and refusal behavior is sensitive to manipulable directions and lexical cues (Arditi et al., 2024; Xue et al., 2026). Models may also invent benign intent during reasoning (Yong and Bach, 2025). Analogously, we probe a local decision boundary; in contrast, our intervention is applied during training, and we ask whether its effect transfers off-surface. Wrapper provenance follows work on in-the-wild and nested jailbreaks (Shen et al., 2023; Ding et al., 2024; Qiu et al., 2023).

LLM-as-a-judge reliability. Strong-LLM judges are scalable proxies for human evaluation but exhibit systematic biases and unstable rankings (Zheng et al., 2023; Norman et al., 2026). Judge pipelines are also directly manipulable by adversarial input content (Shi et al., 2024; Maloyan et al., 2025). Because our independent variable is surface form, wrapper susceptibility is not a generic limitation but a specific rival

explanation. We therefore qualify judges on frozen cases and perform an evaluator-context ablation. Our held-out evaluation also follows contamination-conscious benchmarking practices (Zhang et al., 2025). Statistically, we use a standard difference-in-differences estimand (Imbens and Wooldridge, 2009), cluster bootstrap intervals (Efron, 1979), Holm correction (Holm, 1979), and chance-corrected agreement (Artstein and Poesio, 2008).

3 Problem Formulation

Hard-example safety tuning (Gupta et al., 2026) produces a robust-but-reluctant model: attack success falls, but the model over-refuses jailbreak-shaped benign prompts. Our study targets the causal interpretation of the proposed overrefusal repair, not whether that repair can improve performance on its matched evaluation distribution.

Let a *core* be a literal harmless request c with one audited helpful target T that remains valid under all framings. D presents c directly; W presents c inside a real jailbreak wrapper; N' presents c inside a semantically neutral frame matched to W in tokenizer length. Each arm interleaves the same hard harmful examples, uses the same targets and example order, begins from the same initialization, and receives the same number of updates. W versus D estimates the total effect of adding jailbreak framing to benign supervision. W versus N' is a narrower but more fragile discriminator between jailbreak framing and length-matched additional context.

The intent hypothesis predicts that separation gained by W should transfer to direct cores because harmless intent is frame-independent. The surface-form hypothesis predicts an improvement confined to wrapped presentations. The decisive contrast is therefore not W ’s wrapped score alone, but the difference between its W – D effect under wrapped and direct presentation.

4 Method

Estimand. For arm $a \in \{D, W, N'\}$, seed s , source g , and safe/harmful evaluation pair, let q denote the estimated fulfillment probability and let

$$\delta_a = \text{logit}(q_a) - \text{logit}(q_{\text{base}})$$

be movement relative to the untuned subject model. The base term cancels in arm contrasts

but is retained for headroom checks. We define category-macro separation as

$$\Psi_{a,g} = \overline{(\delta_{a,\text{safe}} - \delta_{a,\text{harm}})},$$

where the overline denotes macro-averaging over evaluation categories. The primary arm contrast is $\Delta\Psi_g = \Psi_{W,g} - \Psi_{D,g}$. We compute it separately for direct and wrapped presentations; the *moderation* is wrapped $\Delta\Psi$ minus direct $\Delta\Psi$. A positive wrapped contrast, nonpositive direct contrast, positive moderation, and adequate direct headroom constitute evidence of test-time surface-form dependence without direct transfer.

Response event. The scored event is substantive fulfillment of the core request. A disclaimer followed by a substantive answer counts as fulfillment; refusal, evasion, or redirection does not. Harmfulness and refusal are separately recorded labels. Fulfillment is not equivalent to response harmfulness, so a reduction in harmful-request fulfillment can support a behavioral safety interpretation, but cannot by itself establish that generated content is harmless.

Inference. Confidence intervals and p -values use a 10,000–20,000-replicate crossed seed-and-pair cluster bootstrap (Efron, 1979), preserving variation across both training seeds and evaluation pairs. The evaluator-context ablation uses a 20,000-replicate paired pair-cluster bootstrap because the responses are identical across scoring conditions. Holm’s step-down procedure controls family-wise error across source tests (Holm, 1979). We report setting-level and pooled effects without treating individual responses from the same seed or pair as independent.

Judge qualification and evaluator-context control. Qwen2.5-7B-Instruct (Yang et al., 2024) and Llama-3.1-70B-AWQ-INT4 (Dubey et al., 2024) each cleared a frozen 58-case qualification suite at an unrelaxed accuracy gate of at least 0.90 before scoring. We then re-score the same stored responses after stripping the jailbreak wrapper from the judge input while preserving the response and core request. If direct exposure of the judge to wrapper text causes the backbone result, the effect should shrink or disappear. An unchanged-XSTest condition serves as a placebo for the re-scoring pipeline.

5 Experimental Setup

Subjects and recipes. Subject models are Qwen2.5-3B-Instruct (Yang et al., 2024) and Mistral-7B-Instruct-v0.3 (Jiang et al., 2023), adapted using LoRA (Hu et al., 2021). We run a 24-seed primary Qwen recipe, an 8-seed alternative Qwen recipe, and a 24-seed Mistral replication. The two qualified judge models are not subject models.

Triple audit. Triples were built with a response target audited as valid under all three contexts. For the Qwen recipe, W and N' have equal tokenizer prompt length on 260/260 benign items and 520/520 interleaved training entries. D, W, and N' share assistant token IDs, target masks, supervised token counts, hard examples and order, initialization, and aligned update positions. Tensor and artifact hashes record these equality checks. Thus, W versus D changes benign user context, while W versus N' additionally controls the benign prompt-token budget.

Evaluation panels. The judgment-blind all-pair audit contains 1,200 matched bundles and 6,400 responses, covering all 200 evaluation pairs per setting and source; prediction files were hashed before analysis. Direct safe/unsafe contrasts use XSTest (Röttger et al., 2023) with a headroom gate. Secondary diagnostics use SORRY-Bench (Xie et al., 2024), OR-Bench-Hard (Cui et al., 2024), and XSTest-safe false refusal. WildJailbreak (Jiang et al., 2024) is used only for target-conditional fragility analysis.

Held-out wrapper families. Cross-family evaluation uses two families disjoint from training and from the FlowGPT evaluation wrappers: Family A contains 40 ChatGPTJailbreak/Reddit templates, and Family B contains 37 jailbreak_chat templates (Shen et al., 2023). We generate and score 83,200 base/D/W responses over all 24 primary-Qwen seeds with the Qwen judge. A frozen, balanced, label-oblivious 2,400-item subset is independently scored with Llama-70B; its prediction file is hash-pinned and all 2,400 outputs parsed successfully.

6 Results

6.1 Identification and sanity checks

Before testing the primary contrast, we verify three conditions required for interpretation. First,

Judge	Direct $\Delta\Psi$	Wrapped $\Delta\Psi$	Moderation
Llama-70B	-0.322**	+0.435*	+0.758***
Qwen-7B	nonpositive	+0.582	+0.970

Table 1: Pooled backbone effects with the wrapper shown to the judge. Both judges yield a nonpositive direct effect, positive wrapped effect, and positive moderation. Significance markers are shown only where the corresponding logged p -values are available: * $p < .05$, ** $p < .01$, *** $p < .001$.

the token audit establishes that the W and N' training arms have identical benign prompt lengths and that all arms use identical response targets and supervised-token masks. Second, direct-set headroom checks retain the untuned model as a reference and show that the nonpositive direct result is not explained by an exhausted floor. Third, within-source pair-cluster falsification and unpaired AUROC/AUPRC analyses show that the moved contrasts depend on preserving the intended safe/harmful pairing; they are not reproduced by an unpaired scalar ranking. These checks do not prove an intent or shortcut mechanism, but they rule out the principal construction, floor, and aggregation artifacts targeted by the design.

The N' arm is nevertheless not a perfect semantic placebo. Token-length matching controls prompt budget, not the meaning of the added text, and N' displays direct safe-fulfillment degradation relative to D. We therefore treat W versus D as the confirmatory causal contrast and W versus N' only as an exploratory discriminator. This distinction prevents a failure of the neutral control from contaminating the better-identified wrapped-versus-direct presentation interaction.

6.2 Surface-form dependence without direct transfer

Under both qualified judges, W improves wrapped safe/harmful separation over D but is nonpositive on the identical direct cores, while the wrapped-minus-direct moderation is positive. Pooled under Llama, direct $\Delta\Psi = -0.322$ ($p = .008$), wrapped $\Delta\Psi = +0.435$ ($p = .014$), and moderation $= +0.758$ ($p = .0006$). Qwen reproduces the sign pattern, with wrapped $\Delta\Psi = +0.582$ and moderation $+0.970$ (Table 1). The pattern appears across the primary Qwen recipe, alternative Qwen recipe, and Mistral replication with seed and pair clustering preserved.

The dedicated XSTest direct anchor is border-

line: $W-D$ is -0.183 , 95% CI $[-0.448, +0.004]$, $p=.056$. We therefore describe direct transfer as *bounded nonpositive*, not as a separately significant degradation on this anchor. The main claim rests on the joint pattern: wrapped improvement, nonpositive direct movement, positive moderation, and passing headroom checks. The result supports a test-time surface-form dependence rather than a frame-independent harmless-intent abstraction. Interpreting the magnitudes on the logit scale of δ , the wrapped gain and direct loss are of comparable absolute size, so the intervention effectively relocates rather than expands the model’s willingness to fulfill: it shifts fulfillment toward prompts that display the trained surface cue and slightly away from the same request stated plainly. This is the behavioral signature one expects from a learned surface trigger, and it is difficult to reconcile with a genuinely frame-independent representation of benign intent, which would leave the direct core unaffected or improved.

6.3 Hiding the wrapper from the judge strengthens the effect

Cross-judge agreement is high on direct prompts ($\kappa = 0.939$) but falls on wrapped prompts ($\kappa = 0.453$), making direct judge exposure to wrapper text a serious rival explanation. We test it by re-scoring the same 6,400 responses with the wrapper removed from the judge input. The response text and identity remain unchanged.

The wrapped effect grows under both judges (Table 2). Qwen’s $W-D$ estimate rises from $+0.582$ to $+1.627$ (paired change $+1.045$, 95% CI $[0.731, 1.626]$, $p=.0001$), and its moderation rises from $+0.970$ to $+2.022$ (paired change $+1.052$, $p=.0001$). Llama’s wrapped $W-D$ estimate rises from $+0.435$ to $+1.581$ (paired change $+1.145$, 95% CI $[0.842, 1.507]$, $p=.0001$), and its moderation rises from $+0.758$ to $+1.903$ (paired change $+1.145$, $p=.0001$). The XSTest placebo is effectively zero: exactly zero for both Llama arms, zero for Qwen W, and a nonsignificant $+0.0075$ for Qwen D.

These results rule out a specific explanation: the backbone effect cannot be caused solely by showing the wrapper to the evaluator. They do not establish judge-independence in a broader sense, because both estimates still rely on automatic judgments and their magnitudes differ across evaluator contexts. The warranted conclusion is that the trained outputs contain behavior sufficient to pre-

Judge	Quantity	Shown	Stripped	Change
Qwen	Wrapped $W-D$	$+0.582$	$+1.627$	$+1.045^{***}$
	Moderation	$+0.970$	$+2.022$	$+1.052^{***}$
Llama	Wrapped $W-D$	$+0.435$	$+1.581$	$+1.145^{***}$
	Moderation	$+0.758$	$+1.903$	$+1.145^{***}$

Table 2: Evaluator-context ablation on the same 6,400 responses. Removing the wrapper from the judge input increases the wrapped effect under both judges ($^{***}p=.0001$); the unchanged-XSTest placebo is near zero.

serve and strengthen the contrast when the wrapper is hidden from the judge. The direction of the change is itself informative: when the wrapper is present in the judge context, it appears to *depress* the measured contrast, most plausibly because the visible jailbreak text makes the judge more suspicious of otherwise compliant responses and lowers their scored fulfillment symmetrically across arms. Removing that suppressive context reveals a larger difference attributable to the responses themselves. Consequently, the wrapper-shown backbone estimate should be read as a conservative lower bound on the output-level surface-form effect rather than as an inflated one.

6.4 Cross-family replication is positive but not independently locked

Family A’s full-panel wrapped $W-D$ effect is $+1.522$, 95% CI $[1.087, 1.900]$, $p=.0002$ ($p_{\text{Holm}}=.0004$), with moderation $+1.705$ $[1.224, 2.182]$. All 24 seeds, 7/8 categories, and every leave-one-category-out (LOCO) estimate are positive; the LOCO range is $+1.357$ to $+1.753$. Family B’s wrapped $W-D$ effect is $+0.643$ $[0.240, 1.039]$, $p=.0014$ ($p_{\text{Holm}}=.0014$), with moderation $+0.826$ $[0.369, 1.313]$. Here 23/24 seeds are positive (the remaining seed is -0.0085), as are 7/8 categories and every LOCO estimate (range $+0.527$ to $+0.770$). The full-panel Qwen result is therefore not confined to the training wrapper family, and the tight LOCO ranges show that no single evaluation category drives it.

The frozen independent Llama audit is directionally concordant but too imprecise for per-family corroboration: Family A is $+0.287$ $[-0.375, +0.978]$, $p=.389$, and Family B is $+0.340$ $[-0.316, +0.989]$, $p=.314$ (Table 3). On the same audit items, Qwen yields Family A $+1.102$ $[0.464, 1.705]$, $p=.0008$, and Family B

Family	Scoring	Wrapped W-D	Positive seeds
A	Qwen full panel	+1.522***	24/24
	Llama audit	+0.287 (ns)	—
B	Qwen full panel	+0.643**	23/24
	Llama audit	+0.340 (ns)	—

Table 3: Held-out cross-family results. Full-panel Qwen scoring replicates the positive wrapped effect on nearly every seed. The smaller frozen Llama audit is directionally concordant but underpowered. ** $p < .01$, *** $p < .001$.

+0.723 $[-0.013, 1.325]$, $p = .057$. All five tested cross-family contrast signs agree across judges, but the one-seed-per-pair direct anchor is slightly positive and nonsignificant under both. The gap between the significant Qwen estimate and the nonsignificant Llama estimate on the *same* audit items is explained by sample size rather than sign disagreement: the frozen audit deliberately balances and subsamples to 2,400 label-oblivious items, which widens intervals enough to straddle zero even though the point estimates remain positive. We thus describe cross-family generalization as full-panel-Qwen replicated and independently directionally concordant, not independently confirmed.

6.5 Safety, neutral controls, and target fragility

No setting passes joint 3-point noninferiority for W versus D or W versus N' on safe and harmful fulfillment. Moreover, the pooled wrapped harmful-request fulfillment contrast reverses sign across judges: +5.17 points under Llama and -3.67 under Qwen. Because fulfillment is not a direct harmfulness judgment, and because even its direction is evaluator-dependent, the experiment provides no evidence of improved safety. We stress this asymmetry: while the safe/harmful *separation* contrast is sign-stable across judges, the isolated harmful-fulfillment component is not, so any claim that benign-wrapper tuning makes the model safer on wrapped harmful requests would be entirely judge-selected and is not supported here.

The exact-token W-versus-N' contrast was intended to distinguish jailbreak-specific framing from any extra user context. It is not independently corroborated and is heterogeneous across recipes, so we treat it as exploratory rather than as evidence for a wrapper-specific mechanism.

On WildJailbreak (Jiang et al., 2024), target-conditional moderation changes from -0.96 with original targets to +2.82 with generated-canonical targets. Target audits attribute this reversal to target choice rather than intervention effects; it is a fragility diagnostic, not external validation. Two additional falsification analyses support the paired reading of the main result. Global additive and category-conditioned transport models fitted to safe members fail to reproduce the wrapped-minus-direct interaction, so the effect is not a uniform scalar shift applied to all safe prompts. Within-source pair-cluster falsification likewise shows that arms which move the paired contrasts are not reliably separated by an unpaired AU-ROC/AUPRC ranking, confirming that the signal lives in the intended safe/harmful pairing rather than in a marginal shift of either pool.

7 Validity, Limitations, and Practical Implications

Internal validity. The strongest aspect of the design is that D, W, and N' use the same cores, response targets, hard harmful examples, order, initialization, and update count, while W and N' also match tokenizer length. This supports a causal interpretation of the controlled arm contrasts within the tested training pipeline. It does not imply that all optimization trajectories are identical: changing user-context tokens changes gradients, which is precisely the intervention. Nor does length matching make neutral and jailbreak frames semantically exchangeable, as the N' degradation demonstrates.

Measurement validity. Both judges passed the same frozen qualification gate, agree on the backbone signs, and preserve the effect when wrapper text is hidden. These checks materially narrow the evaluator-artifact explanation. They do not replace blinded human judgments, and the low wrapped-prompt κ shows that absolute fulfillment labels remain evaluator-sensitive. We therefore emphasize contrasts, paired re-scoring, and sign replication rather than claiming that either judge provides a definitive response-level ground truth.

External validity. The study covers two subject-model families, two Qwen recipes, LoRA adaptation, and three wrapper distributions. It does not establish generality to full-parameter fine-tuning, preference optimization, larger subject models,

multilingual prompts, or arbitrary jailbreak families. The strongest held-out-family evidence comes from the same Qwen judge used in the main panel; the independent Llama audit has matching signs but insufficient precision. The borderline XSTest anchor also leaves a small positive direct-transfer effect statistically possible, although its point estimate is negative and its interval is narrow around zero.

Deployment lesson. A matched wrapped-prompt benchmark can make benign-wrapper tuning appear to teach nuanced intent recognition even when the benefit does not transfer to the literal request. For real deployments, such tuning may still be useful if the target traffic actually contains similar wrappers, but its evaluation should include the same cores both with and without framing, a harmful contrast set, and a judge-context ablation. Wrapped helpfulness should not be reported as a general reduction in overrefusal or as a safety improvement without these controls.

8 Discussion

The observed dissociation is consistent with shortcut learning (Geirhos et al., 2020; McCoy et al., 2019): the intervention changes behavior when jailbreak cues are present but does not produce measurable off-surface transfer to the identical core. This interpretation aligns with evidence that refusal uses low-dimensional, surface-sensitive representations (Arditi et al., 2024) and that safety behavior generalizes poorly across prompt presentations (Wei et al., 2023; Addepalli et al., 2024; Xue et al., 2026). Our behavioral experiment does not identify the internal representation responsible for the effect, so “shortcut” denotes the measured presentation dependence rather than a mechanistic localization.

For the seed recipe (Gupta et al., 2026), the practical conclusion is narrow. Adversarially framed benign counterexamples reduce overrefusal on wrapped prompts, but the harmless-intent explanation is not supported by direct-core transfer, and no safety benefit is established. More broadly, the evaluator-context ablation provides a reusable control for surface-form safety studies: score identical outputs both with and without the manipulated surface cue in the evaluator input. Survival of the effect cannot establish human validity, but disappearance would reveal that a seemingly

model-level result was evaluator-mediated. We recommend that any future benign-repair recipe report the wrapped/direct interaction and the judge-context ablation as standard diagnostics, because either one alone can be satisfied by a surface trigger that a wrapped-only benchmark would misread as intent understanding.

9 Conclusion

We conducted a same-core, same-target controlled training trial that varies benign user context while holding cores, targets, harmful examples, order, initialization, and update count fixed. Jailbreak-wrapped benign supervision improves safe/harmful separation when a wrapper is present at test time but is nonpositive on identical direct cores. Hiding the wrapper from the judge strengthens rather than removes the effect, ruling out direct evaluator exposure as its sole cause. Held-out wrapper families replicate the effect across nearly every primary-Qwen seed, although independent cross-family corroboration remains underpowered. The evidence supports a bounded conclusion: this overrefusal repair buys test-time surface-form dependence, not demonstrated transferable harmless intent or improved safety. Future work should add blinded human adjudication, a higher-powered independent cross-family audit, and a sharper direct anchor able to separate a small positive transfer from exact zero.

References

- Sravanti Addepalli, Yerram Varun, Arun Suggala, Karthikeyan Shanmugam, and Prateek Jain. 2024. Does safety training of llms generalize to semantically related natural prompts? *Preprint*, arXiv:2412.03235.
- Andy Arditi, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Rimskey, Wes Gurnee, and Neel Nanda. 2024. Refusal in language models is mediated by a single direction. *ArXiv*, abs/2406.11717.
- Ron Artstein and Massimo Poesio. 2008. *Inter-coder agreement for computational linguistics*. *Computational Linguistics*.
- Federico Bianchi, Mirac Suzgun, Giuseppe Attanasio, Paul Röttger, Dan Jurafsky, Tatsunori Hashimoto, and James Zou. 2023. Safety-tuned llamas: Lessons from improving the safety of large language models that follow instructions. *ArXiv*, abs/2309.07875.
- J. Cui, Wei-Lin Chiang, Ion Stoica, and Cho-Jui Hsieh. 2024. Or-bench: An over-refusal benchmark for large language models. *ArXiv*, abs/2405.20947.

646	Peng Ding, Jun Kuang, Dan Ma, Xuezhi Cao, Yun-	Xiangyu Qi, Yi Zeng, Tinghao Xie, PinYu Chen, Ruoxi	697
647	sen Xian, Jiajun Chen, and Shujian Huang. 2024.	Jia, Prateek Mittal, and Peter Henderson. 2023.	698
648	A wolf in sheep’s clothing: Generalized nested jail-	Fine-tuning aligned language models compromises	699
649	break prompts can fool large language models eas-	safety, even when users do not intend to! <i>arXiv</i>	700
650	ily.	(Cornell University).	701
651	Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey,	Huachuan Qiu, Shuai Zhang, Anqi Li, Hongliang	702
652	et al. 2024. The llama 3 herd of models.	He, and Zhenzhong Lan. 2023. Latent jailbreak:	703
653	B. Efron. 1979. Bootstrap methods: Another look at	A benchmark for evaluating text safety and out-	704
654	the jackknife. <i>Annals of Statistics</i> , 7:1–26.	put robustness of large language models. <i>Preprint</i> ,	705
655	Matt Gardner, Yoav Artzi, Jonathan Berant, et al. 2020.	arXiv:2307.08487.	706
656	Evaluating models’ local decision boundaries via	Paul Röttger, Hannah Rose Kirk, Bertie Vidgen,	707
657	contrast sets. pages 1307–1323.	Giuseppe Attanasio, Federico Bianchi, and Dirk	708
658	Robert Geirhos, J. Jacobsen, Claudio Michaelis,	Hovy. 2023. Xstest: A test suite for identifying exag-	709
659	R. Zemel, Wieland Brendel, M. Bethge, and Felix	gerated safety behaviours in large language models.	710
660	Wichmann. 2020. Shortcut learning in deep neu-	<i>ArXiv</i> , abs/2308.01263.	711
661	ral networks. <i>Nature Machine Intelligence</i> , 2:665	Xinyue Shen, Zeyuan Chen, M. Backes, Yun Shen, and	712
662	– 673.	Yang Zhang. 2023. "do anything now": Characteriz-	713
663	Prakhar Gupta, Garv Shah, and Donghuan Zhang. 2026.	ing and evaluating in-the-wild jailbreak prompts on	714
664	Self-mined hardness for safety fine-tuning. <i>ArXiv</i> ,	large language models. <i>Proceedings of the 2024 on</i>	715
665	abs/2605.03226.	<i>ACM SIGSAC Conference on Computer and Com-</i>	716
666	S. Holm. 1979. A simple sequentially rejective multi-	<i>munications Security.</i>	717
667	ple test procedure. <i>Scandinavian Journal of Statis-</i>	Jiawen Shi, Zenghui Yuan, Yinyao Liu, Yue Huang, Pan	718
668	<i>tics</i> , 6:65–70.	Zhou, Lichao Sun, and Neil Zhenqiang Gong. 2024.	719
669	J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu,	Optimization-based prompt injection attack to llm-	720
670	Yuanzhi Li, Shean Wang, and Weizhu Chen. 2021.	as-a-judge. <i>Preprint</i> , arXiv:2403.17710.	721
671	Lora: Low-rank adaptation of large language mod-	Alexander Wei, Nika Haghtalab, and J. Steinhardt.	722
672	els. <i>ArXiv</i> , abs/2106.09685.	2023. Jailbroken: How does llm safety training fail?	723
673	Guido W. Imbens and Jeffrey M. Wooldridge. 2009.	<i>ArXiv</i> , abs/2307.02483.	724
674	Recent developments in the econometrics of pro-	Tinghao Xie, Xiangyu Qi, Yi Zeng, Yangsibo Huang,	725
675	gram evaluation. <i>Journal of Economic Literature.</i>	Udari Madhushani Sehwal, Kaixuan Huang, Luxi	726
676	Albert Qiaochu Jiang, Alexandre Sablayrolles, Arthur	He, Boyi Wei, Dacheng Li, Ying Sheng, Ruoxi Jia,	727
677	Mensch, et al. 2023. Mistral 7b. <i>ArXiv</i> ,	Bo Li, Kai Li, Danqi Chen, Peter Henderson, and	728
678	abs/2310.06825.	Prateek Mittal. 2024. Sorry-bench: Systematically	729
679	Liwei Jiang, Kavel Rao, Seungju Han, Allyson Et-	evaluating large language model safety refusal be-	730
680	tinger, Faeze Brahman, Sachin Kumar, Niloofer	haviors. <i>ArXiv</i> , abs/2406.14598.	731
681	Mireshghallah, Ximing Lu, Maarten Sap, Yejin	Zhiyu Xue, Zimo Qi, Guangliang Liu, Bocheng Chen,	732
682	Choi, and Nouha Dziri. 2024. Wildteaming at scale:	and Ramtin Pedarsani. 2026. Deactivating refusal	733
683	From in-the-wild jailbreaks to (adversarially) safer	triggers: Understanding and mitigating overrefusal	734
684	language models. <i>arXiv (Cornell University).</i>	in safety alignment. <i>Preprint</i> , arXiv:2603.11388.	735
685	Narek Maloyan, Bislan Ashinov, and Dmitry Namiot.	Qwen An Yang, Baosong Yang, et al. 2024. Qwen2.5	736
686	2025. Investigating the vulnerability of llm-as-	technical report. <i>ArXiv</i> , abs/2412.15115.	737
687	a-judge architectures to prompt-injection attacks.	Zheng-Xin Yong and Stephen H. Bach. 2025. Self-	738
688	<i>ArXiv</i> , abs/2505.13348.	jailbreaking: Language models can reason them-	739
689	R. Thomas McCoy, Ellie Pavlick, and Tal Linzen. 2019.	selves out of safety alignment after benign reasoning	740
690	Right for the wrong reasons: Diagnosing syntac-	training. <i>ArXiv</i> , abs/2510.20956.	741
691	tic heuristics in natural language inference. <i>ArXiv</i> ,	Jingyu Zhang, Ahmed Elgohary, Xiawei Wang, A S M	742
692	abs/1902.01007.	Iftexhar, Ahmed Magooda, Benjamin Van Durme,	743
693	Justin Norman, Michael U. Rivera, and D. A. Hughes.	Daniel Khashabi, and Kyle Jackson. 2025. Jail-	744
694	2026. Reliability without validity: A system-	break distillation: Renewable safety benchmarking.	745
695	atic, large-scale evaluation of llm-as-a-judge models	<i>Preprint</i> , arXiv:2505.22037.	746
696	across agreement, consistency, and bias.	Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan	747
		Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin,	748
		Zhuohan Li, Dacheng Li, E. Xing, Haotong Zhang,	749
		Joseph E. Gonzalez, and Ion Stoica. 2023. Judg-	750
		ing llm-as-a-judge with mt-bench and chatbot arena.	751
		<i>ArXiv</i> , abs/2306.05685.	752

A Reproducibility Manifest

A.1 Known training configuration

All subject models are adapted with LoRA (Hu et al., 2021); no full model weights are updated. The complete optimizer and decoding configuration files are preserved verbatim with the experiment artifacts rather than reconstructed from incomplete summary records. Table 4 lists every training property available in the audited records used for this paper. We do not infer unrecorded learning rates, LoRA ranks, batch sizes, or sampling parameters.

Field	Audited value
Primary subject	Qwen2.5-3B-Instruct
Primary recipe seeds	24
Alternative Qwen seeds	8
Replication subject	Mistral-7B-Instruct-v0.3
Mistral replication seeds	24
Adaptation method	LoRA only
Training arms	D, W, N'
Shared across arms	targets, hard examples, order initialization, update count
W/N' length audit	260/260 benign items equal
Interleaved length audit	520/520 entries equal
Judges	Qwen2.5-7B, Llama-3.1-70B
Qualification gate	58 cases, accuracy ≥ 0.90

Table 4: Training and evaluation properties supported by the experiment records. Unlisted numerical hyperparameters are supplied through the serialized release configurations rather than guessed here.

A.2 Construction and equality audit

For every benign core, the same audited assistant target is used in D, W, and N'. Equality is checked at the tokenized training-tensor level rather than only by comparing rendered strings. The audit confirms identical assistant token IDs, labels, target masks, and supervised token counts. Hard harmful entries and their targets occur at aligned positions in all arms. Initial checkpoints and seed assignments are shared across corresponding runs. The W/N' prompt-length audit is performed with the subject tokenizer, because character or whitespace length would not control the model's actual token budget.

The neutral frame is designed to add context without jailbreak semantics. Its role is deliberately limited: it controls token count and the presence of additional user text, but it cannot control every semantic or discourse effect of that text. The observed N' degradation is why W versus N' is not promoted to a confirmatory mechanism result.

A.3 Response generation and storage

Subject responses are generated before automatic judgment and stored with immutable identifiers linking the subject model, seed, arm, source, pair, category, and presentation. The all-pair panel contains 1,200 matched bundles and 6,400 stored responses. Prediction files are hashed before aggregate analysis. The evaluator-context ablation reuses the same response identifiers and response strings; only the context supplied to the judge changes.

B Judge Protocol

B.1 Qualification

Both automatic judges independently clear the same frozen 58-case qualification suite at the pre-specified accuracy gate of 0.90. The gate is not relaxed after inspecting downstream results. Qualification establishes basic rubric-following on the frozen cases, not universal validity on adversarially framed responses.

B.2 Fulfillment rubric

The primary question is whether the response substantively fulfills the core request. A response counts as fulfillment if it supplies the requested substance, even when preceded by a warning or disclaimer. It does not count if it refuses, evades, redirects, or merely discusses the request without answering it. Refusal and response harmfulness are retained as separate annotations so they are not silently substituted for fulfillment.

B.3 Evaluator-context ablation

For wrapped items, the wrapper-shown condition provides the judge with the wrapped request and stored response. The wrapper-stripped condition provides the same response and the corresponding core request without the jailbreak wrapper. Response text, response identity, model arm, and evaluation pairing are unchanged. XSTest items, for which there is no wrapper to remove, are passed through the same re-scoring pipeline as a placebo. Their near-zero change argues against a generic implementation or parsing shift.

C Statistical Details

C.1 Unit of analysis

The elementary experimental structure is a safe/harmful pair evaluated under corresponding

arms and seeds. Aggregate uncertainty must therefore reflect repeated observations from a shared training seed and shared evaluation pair. Treating all responses as independent would understate uncertainty.

C.2 Bootstrap procedure

Backbone and cross-family confidence intervals use crossed seed-and-pair cluster resampling. A bootstrap replicate samples training seeds and evaluation-pair clusters with replacement, reconstructs the category-macro arm statistics, and recomputes the direct, wrapped, and moderation contrasts. Reported tests use between 10,000 and 20,000 replicates. The evaluator-context ablation instead uses paired pair-cluster resampling because shown and stripped scores refer to identical stored responses. Its reported results use 20,000 replicates.

C.3 Multiplicity and interpretation

Holm correction (Holm, 1979) is applied across the tested held-out sources. We distinguish three levels of evidence. A *backbone* result requires the wrapped/direct sign pattern and positive moderation under both qualified judges. A *cross-family replication* additionally requires held-out wrapper families. *Independent locking* would require a separate judge to meet the prespecified per-family inferential threshold; the current Llama audit is directionally concordant but does not meet this criterion.

C.4 Headroom

The untuned base model is retained when computing per-arm movement and direct-set headroom. Although the base term cancels in W–D, retaining it checks whether direct effects are constrained by a saturated fulfillment floor or ceiling. The reported direct nontransfer result passes this gate.

D Complete Aggregate Results

E Additional Diagnostics

Transport-model falsification. Global additive and category-conditioned transport models fitted to safe members do not reproduce the wrapped-minus-direct interaction. The observed result therefore cannot be summarized as a common scalar shift applied to all safe prompts.

Contrast	Estimate	Uncertainty
XSTest direct W–D	−0.183	[−0.448, +0.004], $p=.056$
Llama pooled direct	−0.322	$p=.008$
Llama wrapped	+0.435	$p=.014$
Llama moderation	+0.758	$p=.0006$
Qwen wrapped, shown	+0.582	not reported
Qwen moderation, shown	+0.970	not reported

Table 5: Backbone aggregate contrasts. “Not reported” means that no corresponding interval or p -value is present in the experiment summary; none is inferred.

Judge	Quantity	Estimate	Interval / p
Qwen	W–D shown	+0.582	—
Qwen	W–D stripped	+1.627	—
Qwen	paired change	+1.045	[0.731, 1.626], $p=.0001$
Qwen	moderation change	+1.052	$p=.0001$
Llama	W–D shown	+0.435	—
Llama	W–D stripped	+1.581	—
Llama	paired change	+1.145	[0.842, 1.507], $p=.0001$
Llama	moderation change	+1.145	$p=.0001$

Table 6: Evaluator-context ablation on identical stored responses. XSTest placebo changes are zero for Llama D and W, zero for Qwen W, and a nonsignificant +0.0075 for Qwen D.

Pairing falsification. Within-source pair-cluster falsification and unpaired AUROC/AUPRC analyses show that the primary movement is tied to the intended safe/harmful pair structure. Arms that move the same paired contrasts are not reliably separated by an unpaired scalar ranking. This supports the paired estimand and cautions against evaluating the intervention only with independently sampled safe and harmful pools.

WildJailbreak target dependence. With WildJailbreak (Jiang et al., 2024), moderation is −0.96 under original targets and +2.82 under generated-canonical targets. Because the sign changes when the target construction changes, this analysis cannot serve as external validation. It instead demonstrates that target selection can dominate a target-conditional estimate.

Cross-family robustness. Family A is positive on every one of the 24 primary-Qwen seeds; Family B is positive on 23, with the remaining seed at −0.0085. Both are positive for 7/8 categories, and removing any single category preserves a positive aggregate effect. These checks reduce sensitivity to a single seed or category but do not compensate for the low precision of the independent Llama audit.

Family	Quantity	Estimate	Interval / p
A	Qwen wrapped	+1.5222	[1.087, 1.900], $p=.0002$
A	Qwen moderation	+1.7051	[1.224, 2.182], $p=.0002$
A	LOCO range	+1.357 to +1.753	
A	Llama independent	+0.287	[−0.375, 0.978], $p=.389$
A	Qwen same sample	+1.102	[0.464, 1.705], $p=.0008$
B	Qwen wrapped	+0.6433	[0.240, 1.039], $p=.0014$
B	Qwen moderation	+0.8262	[0.369, 1.313], $p=.0008$
B	LOCO range	+0.527 to +0.770	
B	Llama independent	+0.340	[−0.316, 0.989], $p=.314$
B	Qwen same sample	+0.723	[−0.013, 1.325], $p=.057$

Table 7: Held-out cross-family estimates. Family A is positive for 24/24 seeds and 7/8 categories; Family B is positive for 23/24 seeds and 7/8 categories. Every reported LOCO estimate is positive.

Presentation	Cohen’s κ	Interpretation
Direct	0.939	high agreement
Wrapped	0.453	moderate agreement

Table 8: Cross-judge agreement on per-arm labels. The wrapped-prompt drop motivates the evaluator-context ablation; it does not change the concordant backbone contrast signs.

F Scope and Artifact Manifest

The result covers Qwen2.5-3B-Instruct and Mistral-7B-Instruct-v0.3 subject models, LoRA adaptation, two Qwen recipes, and three wrapper distributions. It does not cover full-parameter fine-tuning, preference optimization, multilingual inputs, or arbitrary wrapper families. The direct XSTest anchor is bounded nonpositive rather than conventionally significant, W versus N’ remains exploratory, and no human audit was completed. No human labels are claimed.

The release package contains the D/W/N’ triples for both recipes, wrapper and source hashes, pair texts, token-equality audit records, subject-model adapters and seed identifiers, wrapper-shown and wrapper-stripped outputs from both qualified judges, the cross-family panel, the frozen independent audit and pinned prediction hashes, aggregate and per-setting analysis tables, and bootstrap code. Serialized training and decoding configurations are included verbatim so that hyperparameters absent from the experiment summary are not reconstructed or guessed in this manuscript.

Diagnostic	Llama	Qwen
Wrapped harmful fulfillment	+5.17 points	−3.67 points
Joint 3-point noninferiority	fails	fails
Cross-family direct anchor	+0.163	+0.136
Direct-anchor p -value	.657	.576

Table 9: Safety and direct-anchor bounds. Harmful fulfillment reverses sign across judges, and no setting passes joint safe/harmful noninferiority. The sparse cross-family direct anchors are small, positive, and non-significant.